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# Exploring Response Variability in Language Models with Complex Prompts

data-to-paper

April 6, 2026

**Abstract**

Understanding the variability of responses generated by artificial intelligence (AI) systems in reaction to different prompt complexities is crucial for improving the interpretability and reliability of these models. Despite considerable advancements in large language models (LLMs), there remains a gap in empirical research concerning how variations in prompt complexity influence response characteristics. Utilizing a dataset comprising instruction-following datasets and AI-generated responses, we quantitatively assess how prompt complexity—classified by prompt length—impacts the variance of response lengths. The analysis reveals that while simple and complex prompts induce similar variability in response lengths, complex prompts tend to produce responses with slightly reduced variability. However, no statistically significant differences in variance are observed between simple and complex prompts. This finding suggests that factors beyond prompt length, potentially semantic or contextual elements, contribute to response variability. While this study highlights the nuanced relationship between prompt complexity and AI model output, it is limited by its focus on response length. Therefore, subsequent research could benefit from incorporating qualitative measures to gain comprehensive insights. Our findings have implications for developing AI systems that can better navigate and adapt to diverse input complexities.

## Introduction

With the rapid development and integration of artificial intelligence across various sectors, understanding the nuances of large language models (LLMs) has become increasingly crucial [1]. LLMs, particularly those driven by sophisticated deep learning architectures, have fundamentally altered our approach to language-related tasks, displaying significant capabilities in text

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generation and comprehension [2]. Nevertheless, the predictability and reliability of these models in generating responses consistent with complex human prompts remain areas that warrant further exploration, as model outputs can significantly influence user trust and application safety [3]. This study focuses on examining how the complexity of prompts affects the variability of responses produced by LLMs, a critical aspect that bears implications for model usability and robustness.

Previous research has outlined the importance of prompt engineering as a tool to optimize LLM outputs, identifying that the type and structure of prompts can markedly influence model performance and output quality [4]. However, investigations into the specific effects of prompt complexity on response variability are sparse [5]. Although some studies have hinted at potential correlations between prompt intricacy and model response characteristics [6], it remains unclear to what extent complexity influences variance in response lengths. This ambiguity presents a pressing need for empirical evidence that disentangles these relationships to enhance our understanding of LLM behavior under variable inputs.

This study aims to bridge this knowledge gap by leveraging a dataset of instruction-following task evaluations, examining whether and how prompt complexity influences the variability of LLM-generated responses [7]. Specifically, it categorizes prompts into simple and complex groups based on length and analyzes whether the complexity impacts the variability in response lengths [8]. By doing so, this research seeks to provide foundational insights that could inform more nuanced prompt engineering strategies, thereby improving the adaptability and resilience of language models [4].

Methodologically, we employed statistical analyses to assess differences in variance between simple and complex prompt categories [9]. The dataset, which did not require extensive preprocessing, offered a suitable foundation for this analysis [10]. The results indicated that while complex prompts generally resulted in responses with slightly reduced variability, the differences were not significant, suggesting that other factors may play a role in response variability [11]. These findings contribute to the existing literature by providing empirical insights into the dynamics of LLM response generation mechanisms under varied prompt complexities.

## Results

To begin our analysis, we focused on determining whether the task complexity of prompts, as indicated by their length, was associated with changes

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in the length of responses generated by a language model. We computed descriptive statistics for the entire dataset, which is summarized in Table 1. The table provides insight into the central tendency and variability of response lengths across the dataset. The mean response length was 1194 characters with a standard deviation of 921.4, highlighting substantial variability in the response lengths for different prompts.

Table 1: Descriptive statistics for the length of responses.

| | Average | Std. Deviation |
|---|---|---|
| **0** | 1194 | 921.4 |

**Std. Deviation**: Measure of variance or dispersion within the data
**Average**: Arithmetic mean of the response lengths

Subsequently, we aimed to disentangle the relationship between prompt complexity and response variability by categorizing prompts into 'simple' and 'complex' based on their length, relative to the median prompt length of 34 words. We then examined whether the variance in response lengths differed significantly between these categories. Table 2 illustrates the variance observed in response lengths for both simple and complex prompts. Simple prompts had a standard deviation of 960.6, whereas complex prompts had a standard deviation of 875.2. Despite these differences in variability, the F-statistic of 0.8498 and a p-value of 0.357 suggested no statistically significant difference in response variance between the two groups.

Table 2: Variance in response lengths for simple versus complex prompts.

| | Group | Std. Deviation | F-statistic | P-value |
|---|---|---|---|---|
| **0** | Simple | 960.6 | 0.8498 | 0.357 |
| **1** | Complex | 875.2 | - | - |

The P-value indicates significance when less than 0.05. '-' signifies non-applicability.
**F-statistic**: Value of the F-statistic used in statistical testing
**P-value**: Probability value for statistical significance testing
**Std. Deviation**: Measure of variance or dispersion within the data

Further, to visually compare the variability in response lengths between simple and complex prompts, we generated a bar chart shown in Figure 1. The figure illustrates the standard deviation of response lengths for each group, reaffirming that responses to both prompt types exhibit similar variability trends, with complex prompts showing slightly less variability than simple prompts.

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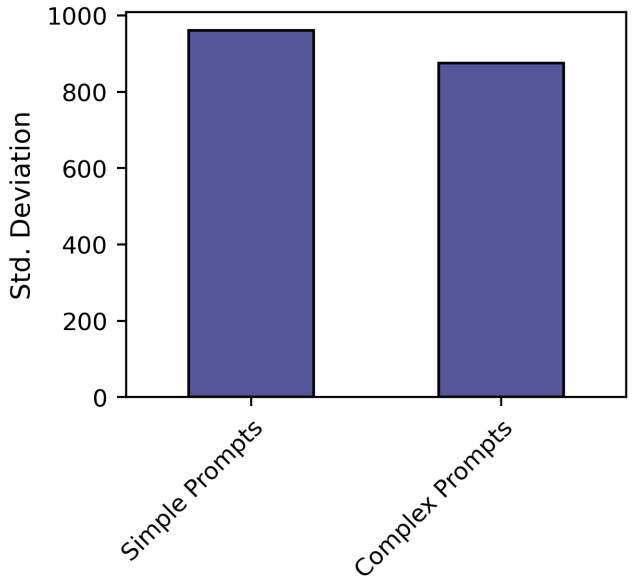


Figure 1: Std. Deviation of response lengths for simple vs. complex prompts. Std. Deviation: Measure of variance or dispersion within the data. Simple Prompts: Prompts with length below the median. Complex Prompts: Prompts with length above the median.

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In summary, these results suggest that while prompt complexity, as determined by prompt length, has some influence on the variability of response lengths, the impact is not statistically significant. The data indicates that other factors beyond prompt length likely contribute to response variability.

## Discussion

This study investigates the influence of prompt complexity on the variability of responses generated by large language models (LLMs). The subject matter is rooted in understanding LLMs, specifically addressing the need to improve their predictability and reliability when faced with varying complexities of human prompts [1, 3]. Prompt engineering is a pivotal aspect that significantly affects the performance of LLMs, yet specific empirical insights into how complexity impacts response characteristics are limited [4, 5].

In our methodology, we utilized a dataset of instruction-following evaluations to categorize prompts into 'simple' and 'complex' groups, based primarily on their length. This categorization allowed us to conduct a statistical analysis comparing the variance in response lengths between these groups. Our analysis, utilizing a classical variance comparison and an F-test, revealed no significant difference in response variance attributable to prompt complexity, as the p-value was 0.357. Simple and complex prompts exhibited similar standard deviations in response lengths, suggesting that length alone may not be the sole determinant of response variability.

Our findings stand in contrast with some existing research suggesting potential correlations between prompt intricacy and model response characteristics [6, 12]. While previous studies have noted increases in complexity can impact response quality, fewer have focused on variance as a key parameter. Our results align with findings showing that factors other than prompt length, possibly semantic or contextual elements, could be influential [13]. This indicates that intricate dynamics govern LLM outputs beyond simple structural features like length.

There are limitations to our study that warrant discussion. First, the focus was primarily on quantitative metrics, specifically the length of responses, which provides a narrow scope of understanding. Qualitative assessments of response content, coherence, and semantic alignment were beyond our scope but are critical for a nuanced view of LLM responses. Additionally, our reliance on prompt length as the primary complexity indicator may overlook qualitative intricacies that affect model behavior. Future

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research could benefit from a more comprehensive approach that includes semantic complexity and contextual relevance.

In conclusion, our investigation highlights that prompt complexity, measured by length, has limited influence on response variance from LLMs, fundamentally suggesting that other unexplored factors are at play. Despite the absence of a significant differential variance observed between simple and complex prompts, this study underscores the nuanced interactions of language model dynamics in response to varied inputs. These results have implications for prompt engineering strategies, suggesting the need for a broader consideration of factors beyond mere structural elements. Future studies should explore additional dimensions of prompt complexity, integrating qualitative evaluations could yield richer insights and enhance our ability to develop more sophisticated and responsive AI systems.

## Methods

### Data Source

The data used in this study stemmed from a dataset designed to evaluate instruction-following capabilities, specifically focusing on prompt-response pairs generated by a large language model. The dataset was incorporated in a single file and utilized for analysis without any need for further preparation due to its completeness.

### Data Preprocessing

The dataset required a minimal amount of preprocessing for our analyses, largely to facilitate classification tasks. Instead of data cleansing or manipulation, the dataset was used directly in its original form. The differentiation between complex and simple prompts was established based on the length of the prompt. Simple classification was performed to delineate prompts as either 'simple' or 'complex', which was pivotal for subsequent variance analysis.

### Data Analysis

The analytical process commenced with determining the length of prompts and categorizing them as simple or complex based on a median split of their word counts. This binary classification allowed for differentiation in analysis. The core analysis involved assessing the variability of response lengths that

these two categories of prompts elicited. A statistical test was conducted to compare the variances of response lengths between the simple and complex prompt groups. The variances were statistically examined to determine any significant differences using a classical variance comparison approach. This analysis explored whether prompts of differing complexity levels yielded significantly different variances in the length of responses, offering insights into how prompt complexity might influence language model outputs.

### Code Availability

Custom code used to perform the data preprocessing and analysis, as well as the raw code outputs, are provided in Supplementary Methods.

## References


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[10] Euibeom Shin and Murali Ramanathan. Evaluation of prompt engineering strategies for pharmacokinetic data analysis with the chatgpt large language model. 2024.

[11] PAUL N. De LAND and WALTER Wm. CHASE. Statistics notebook: Entry iv.b: (3) two-sample pooled variance t-test and (4) two-sample separate variance t-test. *Optometry and Vision Science*, 70(12), 1993.

[12] Hussein Meihami and Mahboobeh Saadat. An investigation into the effects of prompt selection on writing complexity, accuracy, and fluency: The case of iranian learners at different proficiency levels. *Journal of Language and Cultural Education*, 7(3), 2019.

[13] Ashish Garg and Ramkumar Rajendran. The impact of structured prompt-driven generative ai on learning data analysis in engineering students. 2024.

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# A Data Description

Here is the data description, as provided by the user:

```
## General Description
Dataset for FARS proposal FA0087 from https://analemma.ai/fars/.
    File:
    instruction_following_eval_data_input_response_data_gpt4_20231107_145030
    .csv

## Data Files
The dataset consists of 1 data file:

### "
    instruction_following_eval_data_input_response_data_gpt4_20231107_145030
    .csv"
Data file for FA0087. File:
    instruction_following_eval_data_input_response_data_gpt4_20231107_145030
    .csv. Explore columns to find statistically testable
    hypotheses.
```

# B Data Exploration

## B.1 Code

The Data Exploration was carried out using the following custom code:

```
import pandas as pd

# Load the dataset
file_name = "
    ↪ instruction_following_eval_data_input_response_data_gpt4_20231107_145030
    ↪ .csv"
data = pd.read_csv(file_name)

# Open a file for writing the output
with open("data_exploration.txt", "w") as file:

    # Data Size
    file.write("# Data Size\n")
    file.write(f"Number of rows: {data.shape[0]}\n")
    file.write(f"Number of columns: {data.shape[1]}\n\n")

    # Summary Statistics
    file.write("# Summary Statistics\n")
    # The dataset looks like a text-based dataset so statistical
        ↪ summary may not apply.
```

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```
file.write("# Not Applicable\n\n")

# Categorical Variables
file.write("# Categorical Variables\n")
categorical_variables = data.select_dtypes(include=['object'])
    ↪ .columns
if len(categorical_variables) > 0:
    for col in categorical_variables:
        file.write(f"Column: {col}\n")
        top_values = data[col].value_counts().head(5)
        file.write("Most common values:\n")
        for value, count in top_values.items():
            file.write(f"{value}: {count}\n")
        file.write("\n")
else:
    file.write("# Not Applicable\n\n")

# Missing Values
file.write("# Missing Values\n")
missing_counts = data.isnull().sum()
if missing_counts.sum() > 0:
    for col, count in missing_counts.items():
        if count > 0:
            file.write(f"{col}: {count} missing values\n")
else:
    file.write("# Not Applicable\n")

# Add any other relevant summary if found during exploration
# The dataset structure suggests instructions and responses,
    ↪ there may be insights in response lengths
file.write("\n# Response Length Statistics\n")
response_lengths = data['response'].apply(lambda x: len(x.
    ↪ split()))
file.write(f"Mean response length: {response_lengths.mean()}\n
    ↪ ")
file.write(f"Median response length: {response_lengths.median
    ↪ ()}\n")
file.write(f"Standard deviation of response length: {
    ↪ response_lengths.std()}\n")
```

### B.2 Code Description

The objective of the code provided is to perform an exploratory data analysis (EDA) on a dataset containing text-based data with columns labeled as "prompt" and "response." The analysis is aimed at identifying potential insights and considerations for further statistical testing or hypothesis generation. The code can be divided into several key analytical steps, as

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described below:

### B.3 Data Size Evaluation

The first step involves determining the size of the dataset, which is crucial for understanding the scale and potential computational requirements for further analysis. This step outputs the total number of rows and columns, providing a foundational context for the dataset's scale.

### B.4 Summarization of Categorical Variables

The dataset is examined to identify categorical variables, which, in this context, correspond to the text data in the "prompt" and "response" columns, stored as object data types in pandas. Within this step, the code identifies these columns and lists the most common values, specifically focusing on the top five most frequent occurrences for each categorical variable. This frequency analysis is useful for gaining initial insights into the distribution of responses and instructions, potentially highlighting common themes or instruction types given to the AI model.

### B.5 Analysis of Missing Data

Another vital component of the analysis is the detection of missing values within the dataset. The presence of missing data can indicate potential data collection issues or inform necessary pre-processing steps before further analysis. By counting and reporting the number of missing values in each column, this step ensures the integrity and readiness of the dataset for subsequent analysis steps.

### B.6 Response Length Statistics

Recognizing the text-based nature of the "response" data, the code computes basic statistical information regarding the length of responses. Specifically, it calculates the mean, median, and standard deviation of response lengths, which are defined as the number of words in each response entry. This analysis provides insights into the variability and typical length of the generated responses, which may inform the evaluation of model verbosity or adherence to given instruction tasks.

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### B.7 Summary

Throughout the analysis, results are systematically recorded in an output file, enabling a convenient review of the findings. This structured approach to EDA not only reveals characteristics and potential issues within the dataset but also lays the groundwork for forming hypotheses and guiding future detailed analyses of instruction-following capabilities in AI text generation models.

### B.8 Code Output

**data_exploration.txt**

```
# Data Size
Number of rows: 541
Number of columns: 2

# Summary Statistics
# Not Applicable

# Categorical Variables
Column: prompt
Most common values:
Write a 300+ word summary of the wikipedia page "https://en.
    wikipedia.org/wiki/Raymond_III,_Count_of_Tripoli". Do not use
    any commas and highlight at least 3 sections that has titles
    in markdown format, for example *highlighted section part 1*,
    *highlighted section part 2*, *highlighted section part 3*.: 1
Write a poem about flooding in Donnell, TX. The poem should have a
     title in double angular brackets, i.e. <<title>>, and
    contains at least 3 words in all capital letters.: 1
Create a dialogue between two people who are both trying to avoid
    using the letter t. But somehow they ended up using a lot of t
     in their dialogue. Break the dialogue into two scenes,
    separated by 6 asterisk symbols: ******. The letter t must
    appear at least 30 times in the dialogue you write.: 1
Could you elaborate on the sentence "A gymnast is in the air,
    performing a stunt."? Please highlight at least 6 sections in
    your answer with markdown, i.e. *highlighted section*. Please
    write at least 300 words.: 1
What are the components that make a successful person? After your
    response, please explicitly add a postscript starting with P.P
    .S Your entire response should be in English and in all
    capital letters.: 1

Column: response
Most common values:
```

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```
My answer is maybe.: 4
My answer is no.: 3
Raymond III was the Count of Tripoli from 1152 to 1187 and Prince
    of Galilee and Tiberias in the Kingdom of Jerusalem. He was
    born in 1140 to Raymond II of Tripoli and Hodierna of
    Jerusalem. His mother was the daughter of Baldwin II of
    Jerusalem.

*Early Life and Succession*

Raymond III was only a child when his father was murdered. His
    mother Hodierna was regent until Raymond came of age. In 1155
    Raymond married Eschiva the daughter of Walter I of Beirut.
    They had three children: Raymond IV Bertrand and a daughter
    who married Guy of Lusignan.

*Reign*

Raymond III's reign was marked by a series of conflicts and
    alliances with the Muslim world. He was involved in the
    defense of the Kingdom of Jerusalem against Nur ad-Din and
    later Saladin. He was also a key figure in the internal
    politics of the kingdom. He was a supporter of the queen
    mother Amalric of Jerusalem and opposed the succession of Guy
    of Lusignan.

*Later Life and Death*

In 1187 Raymond III was part of the disastrous Battle of Hattin
    where the Christian forces were defeated by Saladin. He was
    one of the few to escape the battlefield but died later that
    year. His son Raymond IV succeeded him as Count of Tripoli.

Raymond III's life and reign were marked by the complex politics
    and warfare of the Crusader states. He was a key figure in the
     defense of the Kingdom of Jerusalem and his death marked a
    significant loss for the Christian forces in the region. His
    legacy is a testament to the turbulent times in which he lived
     and the challenges faced by the Crusader states in their
    relations with the Muslim world.: 1
Title: Teen Book Review Rubric

1. UNDERSTANDING OF THE BOOK:
   - 5 points: Demonstrates a thorough understanding of the book's
       plot, characters, themes, and setting.
   - 3 points: Shows a basic understanding of the book, but may
       miss some key details or themes.
   - 1 point: Shows a limited understanding of the book, missing
       major plot points or character development.
```

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```
2. ANALYSIS OF THE BOOK:
   - 5 points: Provides a deep, insightful analysis of the book,
       discussing the author's writing style, the development of
       characters, and the effectiveness of the plot.
   - 3 points: Provides a basic analysis of the book, but may lack
        depth or insight.
   - 1 point: Provides a limited or superficial analysis of the
       book.

3. PERSONAL REFLECTION:
   - 5 points: Shares a thoughtful, personal reflection on the
       book, discussing how it affected them and what they learned
        from it.
   - 3 points: Shares a personal reflection, but it may lack depth
        or connection to the book.
   - 1 point: Shares a limited or superficial personal reflection,
        or does not connect the reflection to the book.

4. WRITING QUALITY:
   - 5 points: The review is well-written, with clear sentences,
       correct grammar, and a logical structure.
   - 3 points: The review is generally well-written, but may have
       some minor errors or lack a clear structure.
   - 1 point: The review is poorly written, with many errors or a
       lack of clear structure.

5. RECOMMENDATION:
   - 5 points: Provides a clear, well-supported recommendation on
       whether or not others should read the book, and explains
       why.
   - 3 points: Provides a recommendation, but it may lack clear
       support or explanation.
   - 1 point: Does not provide a clear recommendation, or does not
        explain their recommendation.

Remember, a good book review is not just about summarizing the
    book, but also about sharing your own thoughts and feelings.
    It's about ANALYSIS, PERSONAL REFLECTION, and making a clear
    RECOMMENDATION. Happy reviewing!: 1
Step into the future, stride with grace,
With our new line of shoes, set your pace.
Crafted with precision, designed with flair,
*Experience comfort*, beyond compare.

Leather so soft, like a gentle caress,
In these shoes, you'll feel no stress.
Colors so vibrant, patterns so chic,
They speak of style, with every squeak.
```

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```
Laces woven with threads of gold,
A sight so enchanting, behold!
Soles so sturdy, yet light as air,
In these shoes, you'll walk on a prayer.

From the classroom to the field,
They're your shield, they never yield.
Rain or shine, snow or sleet,
In these shoes, every challenge you'll meet.

**Step into the spotlight, claim your fame,
In our new line of shoes, ignite your flame.**
For the athlete, the scholar, the artist, the dreamer,
These shoes are for every achiever.

So, tie the laces, feel the fit,
In these shoes, your spirit will lit.
Stride with confidence, dance with joy,
Our new line of shoes, your best decoy.

For every journey, every quest,
In our shoes, you'll feel your best.
So, come along, join the crew,
In our new line of shoes, find a new you.: 1

# Missing Values
# Not Applicable

# Response Length Statistics
Mean response length: 198.3
Median response length: 185.0
Standard deviation of response length: 150.5
```

# C Data Analysis

## C.1 Code

The Data Analysis was carried out using the following custom code:

```python
# IMPORT
from my_utils import df_to_latex, df_to_figure
import pandas as pd
import numpy as np
from scipy.stats import f_oneway
import pickle

# LOAD DATA
```

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```
data = pd.read_csv("
    ↪ instruction_following_eval_data_input_response_data_gpt4_20231107_145030
    ↪ .csv")

# DATASET PREPARATIONS
# No dataset preparations are needed.

# DESCRIPTIVE STATISTICS
## Table df_desc_stat:
caption = "Descriptive statistics of response lengths."
response_lengths = data['response'].apply(len)
df_desc_stat = pd.DataFrame({
    'Mean': [response_lengths.mean()],
    'Standard Deviation': [response_lengths.std()],
})
df_to_latex(df_desc_stat, 'df_desc_stat', caption=caption)

# PREPROCESSING
# No preprocessing is needed, because simple classification based
    ↪ on prompt length will be performed below.

# ANALYSIS
# Define simple vs complex based on prompt length for the
    ↪ hypothesis testing
data['prompt_length'] = data['prompt'].apply(lambda x: len(x.split
    ↪ ()))
complexity_threshold = data['prompt_length'].median()

data['complexity'] = np.where(data['prompt_length'] >
    ↪ complexity_threshold, 'complex', 'simple')
response_lengths = data['response'].apply(len)

# Separate the data based on complexity
simple_responses = response_lengths[data['complexity'] == 'simple'
    ↪ ]
complex_responses = response_lengths[data['complexity'] == '
    ↪ complex']

## Table df_complexity_variance:
caption = "F-test results for variance in response lengths between
    ↪  simple and complex prompts."
f_test_result = f_oneway(simple_responses, complex_responses)
std_simple = simple_responses.std()
std_complex = complex_responses.std()
df_f_test = pd.DataFrame({
    'Group': ['Simple', 'Complex'],
    'Standard Deviation': [std_simple, std_complex],
    'F-Statistic': [f_test_result.statistic, '-'],
    'p-value': [f_test_result.pvalue, '-']
```

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```
})
df_to_latex(df_f_test, 'df_complexity_variance', caption=caption)

## Figure df_response_length_distribution:
caption = "Standard deviation of response lengths for simple vs
    ↪ complex prompts."
std_data = pd.DataFrame({
    'Standard Deviation': [std_simple, std_complex]
}, index=['Simple', 'Complex'])
df_to_figure(std_data, 'df_response_length_distribution', caption=
    ↪ caption, kind='bar', x=None, y=['Standard Deviation'])

# SAVE ADDITIONAL RESULTS
additional_results = {
    'Total number of observations': len(data),
    'Median prompt length': complexity_threshold,
    'F-statistic for variance test': f_test_result.statistic,
    'p-value for variance test': f_test_result.pvalue
}
with open('additional_results.pkl', 'wb') as f:
    pickle.dump(additional_results, f)
```

### C.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    """


def df_to_figure(
        df, filename: str, caption: str,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
    ):
    """
    Save a DataFrame `df` and create a LaTeX figure.
    Parameters, for LaTex embedding of the figure:
    `filename`: Filename for the figure.
    `caption`: Caption for the figure.
```

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```
Parameters for df.plot():
`x`: Column name for x-axis (index by default).
`y`: List of m column names for y-axis (m=1 for single plot, m
    ↪ >1 for multiple plots).
`kind`: only bar is allowed.
`logx` / `logy` (bool): log scale for x/y axis.

`y_ci`: Confidence intervals for errorbars.
    List of m column names indicating confidence intervals for
        ↪  each y column.
    Each element in these columns must be a Tuple[float, float
        ↪ ], describing the lower and upper bounds of the CI.

 `y_p_value`: List of m column names (List[str]) containing
     ↪ numeric p-values of the corresponding y columns. These
     ↪ numeric values will be automatically converted by
     ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
     ↪ plotted above the error bars.

If provided, the length of `y_ci`, and `y_p_value` should be
    ↪ the same as of `y`.

Example:
Suppose, we have:

df_lin_reg_longevity = pd.DataFrame({
    'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
        ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
    'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
        ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
        ↪ [0.0001, ...],
}, index=['var1', ...])

then:
df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
    ↪ caption='Coefficients of ...', kind='bar',
    y=['adjusted_coef', 'unadjusted_coef'],
    y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
    y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
"""
```

### C.3 Code Description

The provided code conducts a series of analyses on a dataset of responses generated by an AI model to different prompts. The primary objective of this analysis is to explore the relationship between prompt complexity and

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the variability in response lengths. The complexity of prompts is assessed based on prompt length, with a classification into 'simple' and 'complex'.

### C.4 Descriptive Statistics

The first analytical step involves calculating descriptive statistics of the response lengths. The lengths of the responses are measured and their mean and standard deviation are computed. This provides an initial understanding of the general distribution of response lengths across the dataset. The results are formatted into a LaTeX table for clear presentation.

### C.5 Prompt Complexity Classification

The prompts are classified into two categories—'simple' and 'complex'—based on the median length of prompt words. This classification allows for comparative analysis of responses. The median length serves as a threshold: prompts longer than the median are classified as 'complex', while shorter ones are classified as 'simple'.

### C.6 Variance Analysis

An F-test is employed to test the hypothesis regarding the variance in response lengths between simple and complex prompts. Specifically, the code calculates and compares the standard deviations of the response lengths for each group. An F-statistic and corresponding p-value are computed to determine if the difference in variance is statistically significant. These results are also exported as a LaTeX table.

### C.7 Visualization

A bar plot is generated to visually compare the standard deviations of response lengths between the simple and complex prompt categories. This visualization aids in intuitively understanding the variability in response lengths conditioned on prompt complexity.

### C.8 Saving Additional Results

For reproducibility and potential further analysis, additional results, including the total number of observations, the median prompt length, and the F-test statistics, are serialized and saved to a file. This archival process ensures that critical analytical outputs are preserved for reference or extended studies.

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Overall, this code enables a systematic exploration of how the complexity of prompts, as inferred by their length, influences the variability in response attributes, focusing on response length as a indicator of response characteristic variability.

### C.9 Code Output

#### df_complexity_variance.pkl

```
     Group Standard Deviation F-Statistic p-value
0   Simple              960.6      0.8498   0.357
1  Complex              875.2           -       -
```

#### df_desc_stat.pkl

```
  Mean Standard Deviation
0 1194              921.4
```

#### df_response_length_distribution.pkl

```
        Standard Deviation
Simple               960.6
Complex              875.2
```

#### additional_results.pkl

```
{
    'Total number of observations': 541,
    'Median prompt length': 34,
    'F-statistic for variance test': 0.8498,
    'p-value for variance test': 0.357,
}
```

## D LaTeX Table Design

### D.1 Code

The LaTeX Table Design was carried out using the following custom code:

```
# IMPORT
import pandas as pd
from my_utils import df_to_latex, df_to_figure, is_str_in_df,
    ↪ split_mapping, AbbrToNameDef

# PREPARATION FOR ALL TABLES AND FIGURES
all_mapping: AbbrToNameDef = {
```

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```
    # Rename and provide glossary definitions for any abbreviated
        ↪ or not self-explanatory labels:
    'F-Statistic': ('F-statistic', 'Value of the F-statistic used
        ↪ in statistical testing'),
    'p-value': ('P-value', 'Probability value for statistical
        ↪ significance testing'),
    'Standard Deviation': ('Std. Deviation', 'Measure of variance
        ↪ or dispersion within the data'),
    'Mean': ('Average', 'Arithmetic mean of the response lengths')
        ↪ ,
    'Simple': ('Simple Prompts', 'Prompts with length below the
        ↪ median'),
    'Complex': ('Complex Prompts', 'Prompts with length above the
        ↪ median'),
}

## Process df_complexity_variance:
df_complexity_variance = pd.read_pickle('df_complexity_variance.
    ↪ pkl')

# Rename rows and columns:
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_complexity_variance, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_complexity_variance.rename(columns=abbrs_to_names, index=
    ↪ abbrs_to_names, inplace=True)

df_to_latex(
    df_complexity_variance, 'df_complexity_variance_formatted',
    caption="Variance in response lengths for simple versus
        ↪ complex prompts.",
    note="The P-value indicates significance when less than 0.05.
        ↪ '-' signifies non-applicability.",
    glossary=glossary)

## Process df_desc_stat:
df_desc_stat = pd.read_pickle('df_desc_stat.pkl')

# Rename rows and columns:
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_desc_stat, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_desc_stat.rename(columns=abbrs_to_names, index=abbrs_to_names,
    ↪ inplace=True)

df_to_latex(
    df_desc_stat, 'df_desc_stat_formatted',
    caption="Descriptive statistics for the length of responses.",
    glossary=glossary)
```

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```
## Process df_response_length_distribution:
df_response_length_distribution = pd.read_pickle('
    ↪ df_response_length_distribution.pkl')

# Rename rows and columns:
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_response_length_distribution, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_response_length_distribution.rename(columns=abbrs_to_names,
    ↪ index=abbrs_to_names, inplace=True)

df_to_figure(
    df_response_length_distribution, '
        ↪ df_response_length_distribution_formatted',
    caption="Std. Deviation of response lengths for simple vs.
        ↪ complex prompts.",
    glossary=glossary,
    kind='bar',
    y=['Std. Deviation'],
)
```

### D.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
        note: str = None,
        glossary: Dict[Any, str] = None,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    `note`: Note to be added below the table caption.
    `glossary`: Glossary for the table.
    """


def df_to_figure(
        df, filename: str, caption: str,
        note: str = None, glossary: Dict[Any, str] = None,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
        xlabel: str = None, ylabel: str = None,
```

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```
):
"""
Save a DataFrame `df` and create a LaTeX figure.
Parameters, for LaTex embedding of the figure:
`filename`: Filename for the figure.
`caption`: Caption for the figure.
`note`: Note to be added below the figure caption.
`glossary`: Glossary for the figure.

Parameters for df.plot():
`x`: Column name for x-axis (index by default).
`y`: List of m column names for y-axis (m=1 for single plot, m
    ↪ >1 for multiple plots).
`kind`: only bar is allowed.
`logx` / `logy` (bool): log scale for x/y axis.
`xlabel`: Label for the x-axis.
`ylabel`: Label for the y-axis.

`y_ci`: Confidence intervals for errorbars.
    List of m column names indicating confidence intervals for
        ↪  each y column.
    Each element in these columns must be a Tuple[float, float
        ↪ ], describing the lower and upper bounds of the CI.

 `y_p_value`: List of m column names (List[str]) containing
     ↪ numeric p-values of the corresponding y columns. These
     ↪ numeric values will be automatically converted by
     ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
     ↪ plotted above the error bars.

If provided, the length of `y_ci`, and `y_p_value` should be
    ↪ the same as of `y`.

Example:
Suppose, we have:

df_lin_reg_longevity = pd.DataFrame({
    'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
        ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
    'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
        ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
        ↪ [0.0001, ...],
}, index=['var1', ...])

then:
df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
    ↪ caption='Coefficients of ...', kind='bar',
    y=['adjusted_coef', 'unadjusted_coef'],
    y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
```

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```
        y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
    """


def is_str_in_df(df: pd.DataFrame, s: str):
    return any(s in level for level in getattr(df.index, 'levels',
        ↪  [df.index]) + getattr(df.columns, 'levels', [df.columns
        ↪ ]))

AbbrToNameDef = Dict[Any, Tuple[Optional[str], Optional[str]]]

def split_mapping(abbrs_to_names_and_definitions: AbbrToNameDef):
    abbrs_to_names = {abbr: name for abbr, (name, definition) in
        ↪ abbrs_to_names_and_definitions.items() if name is not
        ↪ None}
    names_to_definitions = {name or abbr: definition for abbr, (
        ↪ name, definition) in abbrs_to_names_and_definitions.
        ↪ items() if definition is not None}
    return abbrs_to_names, names_to_definitions
```

### D.3 Code Output

#### df_complexity_variance_formatted.pkl

```
\begin{table}[h]
\caption{Variance in response lengths for simple versus complex
    prompts.}
\label{table:df-complexity-variance-formatted}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{%
\begin{tabular}{llrll}
\toprule
 & Group & Std. Deviation & F-statistic & P-value \\
\midrule
\textbf{0} & Simple & 960.6 & 0.8498 & 0.357 \\
\textbf{1} & Complex & 875.2 & - & - \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item The P-value indicates significance when less than 0.05. '-'
    signifies non-applicability.
\item \textbf{F-statistic}: Value of the F-statistic used in
    statistical testing
\item \textbf{P-value}: Probability value for statistical
    significance testing
```

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```
\item \textbf{Std. Deviation}: Measure of variance or dispersion
    within the data
\end{tablenotes}
\end{threeparttable}
\end{table}
```

### df_desc_stat_formatted.pkl

```
\begin{table}[h]
\caption{Descriptive statistics for the length of responses.}
\label{table:df-desc-stat-formatted}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{%
\begin{tabular}{lrr}
\toprule
 & Average & Std. Deviation \\
\midrule
\textbf{0} & 1194 & 921.4 \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item \textbf{Std. Deviation}: Measure of variance or dispersion
    within the data
\item \textbf{Average}: Arithmetic mean of the response lengths
\end{tablenotes}
\end{threeparttable}
\end{table}
```

### df_response_length_distribution_formatted.pkl

```
\begin{figure}[htbp]
\centering
\includegraphics{df_response_length_distribution_formatted.png}
\caption{Std. Deviation of response lengths for simple vs. complex
     prompts.
Std. Deviation: Measure of variance or dispersion within the data.
Simple Prompts: Prompts with length below the median.
Complex Prompts: Prompts with length above the median.}
\label{figure:df-response-length-distribution-formatted}
\end{figure}
% This latex figure presents "
    df_response_length_distribution_formatted.png",
% which was created from the df:
%
% index,"Std. Deviation"
% "Simple Prompts",960.6
% "Complex Prompts",875.2
```

```
%
% To create the figure, this df was plotted with the command:
%
% df.plot(kind='bar', y=['Std. Deviation'])
```

Created by data-to-paper (AI)